



Graphene-CoO/PEG composite phase change materials with enhanced solar-to-thermal energy conversion and storage capacity

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ABSTRACT

Composite phase change materials (PCMs) with excellent light adsorption and high thermal conductivity have great potentials in the efficient solar-to-thermal energy utilization. In order to improve the light absorption and solar-to-thermal conversion efficiency, reduced graphene oxide nanosheets decorated with CoO nanoparticles (rGO@Co) were used as support for polyethylene glycol (PEG) PCMs. The hierarchical structure composed of rGO nanosheets provides a large space to impregnate a high amount of PEG with good shape stability during the solid-liquid phase transition. Meanwhile, CoO nanoparticles and rGO nanosheets with high light absorption can provide rGO@Co with enhanced light absorption capacity and solar-to-thermal conversion capacities. In addition, the well distributed CoO nanoparticles on rGO nanosheets could contact PEG chains better, act as heat transfer intermediate and reduce the interface thermal resistance for the thermal energy transfer between rGO and PEG during the thermal energy storage/release process. Therefore, the prepared rGO@Co/PEG composite PCMs exhibit enhanced thermal conductivity with 51.79%~152.46% increase as compared with pure PEG, high energy storage densities (131.94 J/g), good shape stability and high solar-to-thermal energy storage/release rates. Hence, the light-driven composite PCMs have great potentials for the efficient utilization of sustainable solar energy.

1. Introduction

In the past decades, various kinds of energy storage and conversion technologies have been researched and developed to alleviate the energy crisis, such as using sustainable solar energy (e.g., photocatalytic water splitting [1,2], photocatalytic CO₂ reduction [3], solar cells [4], solar-to-thermal conversion [5,6]), increasing the efficiency of fossil fuels [7], or utilizing phase change materials (PCMs) to store/release the waste heat [8]. Among these approaches, solar-to-thermal conversion has been attracting increasing attention and widely studied by researchers due to their environmental friendliness, cost effectiveness, and great potentials in practical applications [9,10]. In the solar-to-thermal conversion, the light-absorbing medium converted the energy from absorbed light into thermal energy. Therefore, the efficiency of solar-to-thermal conversion is strongly influenced by the light-absorbing medium, the solar-to-thermal conversion rate and the thermal energy storage materials.

Different from the traditional sensible heat storage materials, PCMs could realize high thermal storage and release capacity during the solid-liquid phase change process, as well as small temperature variation and excellent stability. As a result, PCMs have been attracting increasing attention and applied in many fields, such as waste heat recovery, intelligent air-condition buildings, solar energy conversion, thermo-regulated textiles and so on [11]. To prevent the leakage during the solid-liquid phase change process, numerous solid supports (e.g., graphene oxide [12], graphene aerogel [13,14], carbon nanotube array [15,16], mesoporous silica [17], carbon aerogel [18], metal organic frameworks [19], expanded graphite [20,21]) have been investigated to encapsulate the PCMs in the shell layer or the porous channels as shape-stabilized composite PCMs [22–24]. However, for the solar-to-thermal conversion, carbon nanomaterials (e.g., expanded graphite, graphene nanosheets, carbon nanotubes, porous graphitized carbon) with high surface area and thermal conductivity, to encapsulate PCMs has been found to be one of the most effective strategies, which

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can not only realize the solar energy capture and solar-to-thermal conversion, but also store high thermal energy. Previous studies have demonstrated that reduced graphene oxide (rGO), carbon nanotube and carbon fiber as fillers or supports in PCMs composite can increase the photothermal conversion efficiency [25,26]. Nevertheless, the solar-to-thermal conversion efficiency of carbon-based PCMs is still not high enough due to the limitation of the thermal conductivity of the supports. Thus, the design of carbon-based PCMs with higher solar-to-thermal conversion and thermal storage efficiency is an indispensable way to alleviate the increasingly severe energy crisis [27].

As well known, one of the popular strategies to enhance the thermal conductivity of composite PCMs is to introduce highly thermally conductive metal particles into the PCMs system. Qian et al. prepared a series of highly thermally conductive composite PCMs by blending diatomite modified with Ag nanoparticles and PEG. The results showed that the phase change enthalpy of PEG/diatomite/Ag PCMs reached 111.3 J/g, and the thermal conductivity of PEG/diatomite PCMs containing 7.2 wt% Ag was $0.82 \text{ W m}^{-1} \text{ K}^{-1}$, which was higher than that of PEG/diatomite composite [28,29]. In addition to precious metals, magnetic nanoparticles (e.g., Fe_3O_4) have also been used to prepare highly thermally conductive composite PCMs. Wang et al. prepared magnetocaloric Fe_3O_4 -modified graphene nanosheets (Fe_3O_4 -GNS) with improved light-collecting properties, which was then uniformly dispersed in PEG/ SiO_2 PCMs with efficient electromagnetic and solar energy conversion and storage [30,31]. At present, the application of transition metal or oxide nanoparticles on the solar-to-thermal conversion of composite PCMs is still scarce. Among the known transition metal oxides, diamond oxides (e.g., CoO) have been widely considered as having the best thermal transfer performance, but they have never been applied to the field of phase change.

In this work, rGO nanosheets decorated with well-distributed CoO nanoparticles were used as polyethylene glycol (PEG) support to improve the solar-to-thermal conversion for the first time, in which CoO nanoparticles act as heat transfer medium and reduce the interface thermal resistance between rGO and PEG chains. Besides, due to the high light absorption capacity and excellent thermal conductivity of rGO and CoO [32,33], rGO@Co improved the solar-to-thermal energy conversion efficiencies of the rGO@Co/PEG composite PCMs. What's more, the prepared rGO/PEG and rGO@Co/PEG composite PCMs also have improved thermal conductivities, increased solar-to-thermal conversion rates, excellent shape stabilities and high thermal energy storage efficiencies.

2. Experimental

2.1. Raw materials

Graphite flake (AR), KMnO_4 (AR), 30 wt% H_2O_2 (AR), NaNO_3 (AR), 36.5% HCl (AR) and concentrated H_2SO_4 (AR) were purchased from Sinopharm Chemical Reagent Co., Ltd. NaOH (98%), $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ (98.0–102.0%) and polyethylene glycol 2000 (PEG2000) were purchased from Alfa Aesar Ltd.

2.2. Synthesis of rGO@Co composites

Graphene oxide (GO) was prepared from graphite flakes by the modified Hummers' method [34]. Typically, 1 g of graphite and 0.5 g of NaNO_3 were dispersed into 40 ml of concentrated sulfuric acid by continuous stirring under ice water bath. Then, 3 g of KMnO_4 was slowly added to the above solution and stirred in ice water bath for 2 h. After continuing stirring at room temperature for 12 h, 100 ml of boiling water was added into the above mixture and stirred for another 20 min. 7 ml of H_2O_2 aqueous solution (30 wt%) was added under continuous stirring until no more bubbles were formed. The graphene oxide nanosheets were collected by centrifugation, washed several times with 5% HCl solution and deionized water until neutral, followed by freeze

drying.

The rGO@Co composites decorated with CoO nanoparticles were prepared via a thermal annealing method according to our previous results [35], as shown in Scheme 1a. Typically, 48 mg of GO was well dispersed in 24 ml of deionized water under sonication for 2 h. Then a certain amount (6 mg, 8 mg or 12 mg, respectively) of $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ was added into the GO solution and stirred at room temperature for 12 h. After adjusting the pH of the solution to ~ 11 with 1 M NaOH and stirring for 12 h, the resulting solid was collected, washed with water several times and freeze-dried for 24 h. The collected black powder was pyrolyzed directly at 500°C for 2 h under flowing N_2 atmosphere. According to the mass ratio of GO to $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, the obtained products were named as rGO@Co-8, rGO@Co-6, and rGO@Co-4 in the order of increased CoO content in the composites. As a control, rGO was prepared using GO under the same pyrolysis conditions.

2.3. Preparation of rGO@Co/PEG composites

rGO@Co/PEG composites with 80 wt% PEG2000 were prepared by a physical mixing and melt impregnation method (Scheme 1b). Prior to preparing the composite PCMs, the rGO and rGO@Co were vacuumed at 100°C for 24 h to remove the possible gases/ H_2O . Then 0.4 g of rGO, rGO@Co-8, rGO@Co-6 and rGO@Co-4 were introduced separately into 20 ml of ethanol containing 1.6 g of PEG2000. The mixture was sonicated for 30 min, then vigorously stirred at 90°C for 1 h, and then vacuum-dried at 80°C for 24 h until the ethanol solution was completely evaporated. The prepared composites were named as rGO/PEG, rGO@Co-8/PEG, rGO@Co-6/PEG and rGO@Co-4/PEG, respectively.

2.4. Characterization

The morphology of the samples was checked by field emission scanning electron microscopy (FESEM, ZEISS SUPRA55) and transmission electron microscopy (TEM, TEI Tecnai F20). The phase structure of the samples was characterized by X-ray diffractometry (XRD, M21X) using Cu K α ray ($\lambda = 1.541 \text{ \AA}$). A Fourier transform infrared spectroscopy (FTIR, Nicolet 6700) was used to determine the chemical functional groups. The thermal stability was evaluated using a synchronous thermal analyzer (TG-DSC, Netzsch STA449F3) under flowing nitrogen atmosphere. Raman spectroscopy (FINDER Vista) was performed to measure the graphitization of porous carbon. Differential scanning calorimeter (DSC, Q2000) measurement was used to characterize the thermal properties of composite PCMs. The thermal conductivity of composite PCMs was measured by a laser flash system (Laser Flash Apparatus NETZSCH MicroFlash LFA 457) at room temperature.

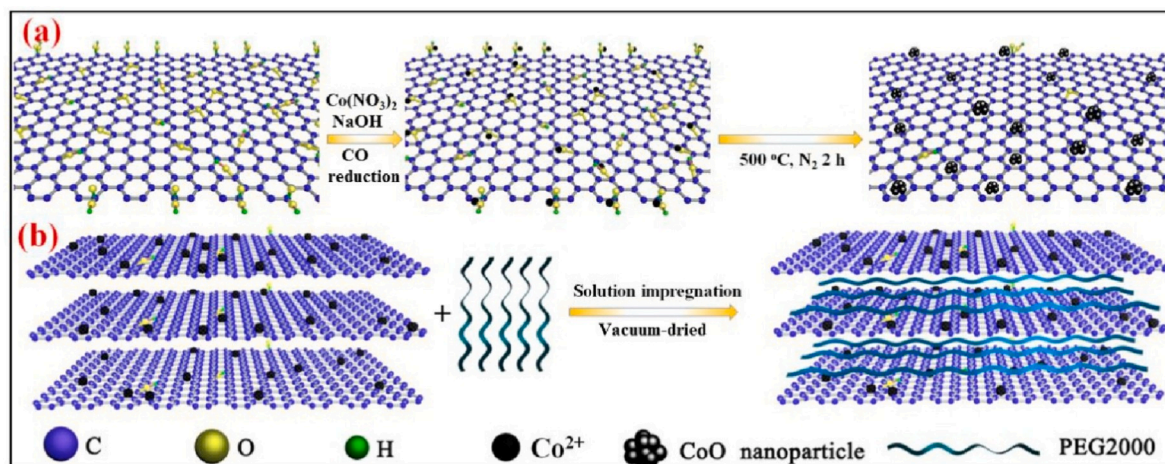
2.5. Solar-to-thermal conversion test

The experimental device of the solar-to-thermal conversion system includes light capture, solar-to-thermal conversion, data collection and processing equipment. Solar-to-thermal conversion tests were performed on PCMs under simulated sunlight using a 300 W xenon lamp. First, 0.12 g of PCMs sample was ground and pressed into a tablet under a pressure of 8 MPa. Secondly, the pressed sample is placed in a position where the light intensity is about one solar light intensity for testing. The infrared temperature probe is used to monitor the temperature of the target sample at the same time. The data acquisition system with Temp Monitor S2 software was used to collect the data with the solar-to-thermal conversion process.

3. Results and discussion

3.1. Morphology and structure of prepared rGO@Co composites

As characterized by SEM, TEM, XRD and FT-IR (Fig. S1), GO nanosheets have been successfully synthesized. In order to investigate the



Scheme 1. The synthesis route of rGO@Co (a) and rGO@Co/PEG composites (b).

effect of CoO amount on the solar-to-thermal conversion and storage performance, the CoO content in rGO@Co was regulated by adjusting the initial $\text{Co}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ amount. The SEM images of rGO, rGO@Co-8, rGO@Co-6 and rGO@Co-4 composites are shown in Fig. 1. As can be seen, rGO displays a smooth surface and hierarchical structure composed of nanosheets (Fig. 1a). For rGO@Co-8, CoO nanoparticles with different particle sizes are sparsely located on the rGO surface (Fig. 1b). With increasing Co amount (e.g., for rGO@Co-6), CoO nanoparticles with 10–100 nm particle size are evenly distributed on the rGO surface and/or embedded in the rGO layers (Fig. 1c). For rGO@Co-4 with more Co amount, the CoO nanoparticles with around 20–100 nm in size and well distribution are embedded in/on the whole surface of the rGO nanosheets (Fig. 1d). As shown in the inset SEM images (Fig. 1), rGO@Co composites maintain the nanosheet and hierarchical morphology of rGO. The TEM and EDS analysis of rGO@Co-6 are shown in Fig. 2. The EDS result (Fig. 2b) demonstrates shows the existence of C, O and Co while the elemental distribution images of Co, O and C (Fig. 2a) indicates the well distribution of CoO nanoparticles. The whole coverage of O elements on rGO@Co-6 may be due to the existence of oxygen containing groups and chemically adsorbed gases (e.g., CO_2 , O_2) besides the CoO nanoparticles. In Fig. 2c, CoO nanoparticles are well

distributed on the rGO nanosheets due to the strong electrostatic interaction between Co^{2+} and oxygen-containing groups on GO to fix Co^{2+} species, which can in-situ form $\text{Co}(\text{OH})_2$ during precipitation by NaOH and further CoO nanoparticles during thermal treatment under flowing N_2 at 500°C . In the HRTEM image (Fig. 2d), the lattice spacings of 0.20 nm and 0.24 nm respectively correspond to the (200) and (111) planes belonging to CoO phase, indicating the existence of CoO phase [36].

As displayed in Fig. S1, the peak at 10.2° corresponds to the C(001) lattice of highly oxidized GO with an interlayer spacing of 0.85 nm. After thermal reduction, the rGO showed a broad diffraction peak at 15° – 30° (Fig. 3a), revealing the disordered stacking of the rGO and the successful synthesis of rGO from GO via the thermal reduction method. The small diffraction peaks at 37.9° and 44.3° in the rGO@Co XRD pattern (Fig. 3a) correspond to the (111) and (200) crystal planes of the rocksalt-phase CoO crystal (JCPDS 43–1004), respectively while the peaks at 30.6° , 34.4° , 37.9° and 47.8° can be indexed to the (100), (002), (101) and (102) crystal planes of the wurtzite CoO crystal (JCPDS 80–0075) [36,37], indicating the formation of CoO upon thermal treatment process. The CoO nanoparticles were in-situ formed between rGO nanosheets during the thermal treatment and reduced the aggregation of the rGO nanosheets, leading to the weakened diffraction intensity of the

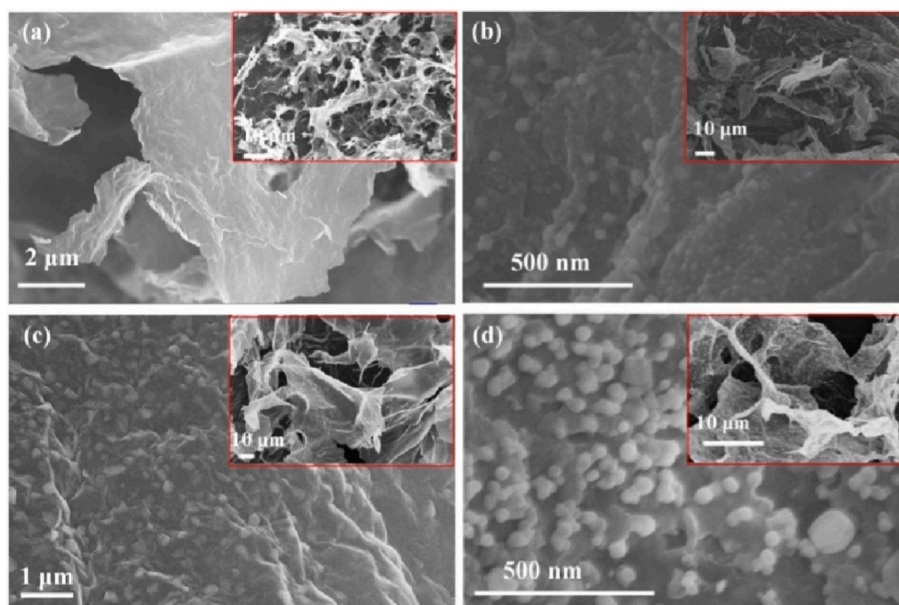


Fig. 1. SEM images of rGO (a), rGO@Co-8 (b), rGO@Co-6 (c) and rGO@Co-4 (d).

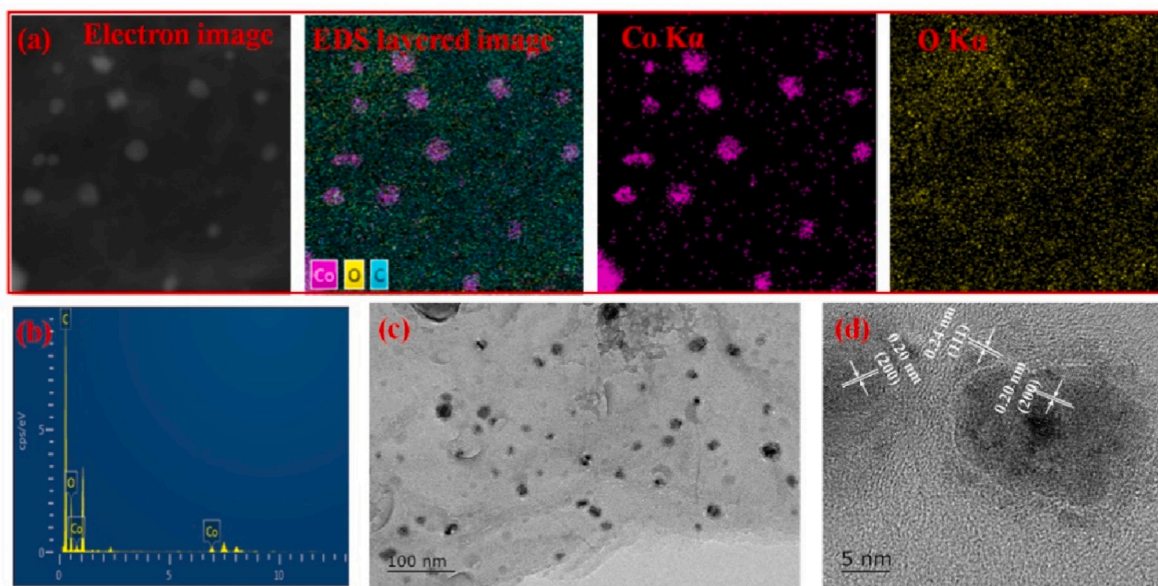


Fig. 2. EDS elemental mapping images (a), EDS analysis (b), TEM (c) and HRTEM (d) images of rGO@Co-6.

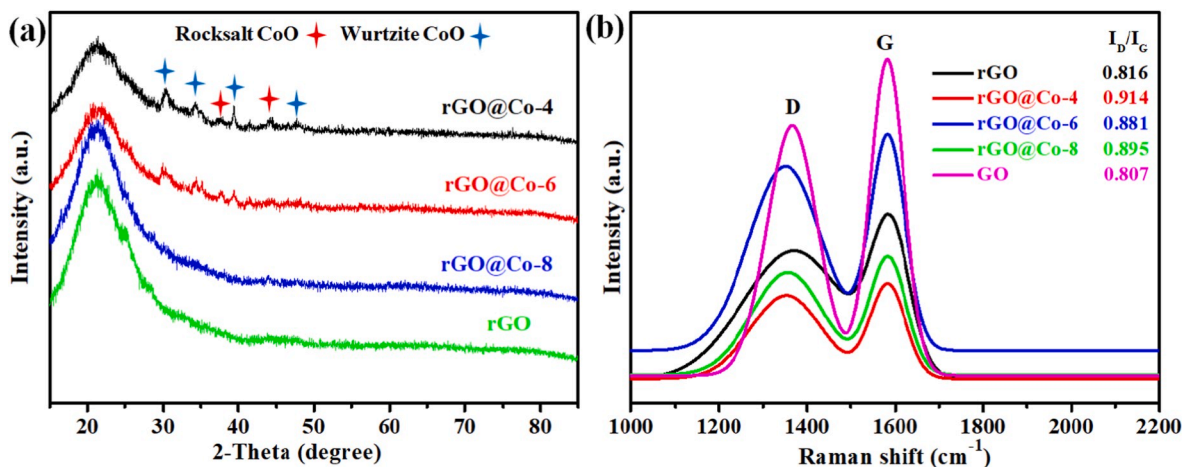


Fig. 3. (a) XRD patterns of rGO and rGO@Co composites; (b) Raman spectra of GO, rGO and rGO@Co composites.

characteristic broad peak of graphene with the increased CoO amount [38]. Raman spectroscopy is widely used to check the structural properties of graphene-based materials, such as structural defects and disorders. The Raman spectra of GO, rGO and rGO@Co composites was conducted, as shown in Fig. 3b. It is obvious that two prominent peaks at around 1350 cm^{-1} (D band) and 1600 cm^{-1} (G band) were detected, in which the D band is associated with structural defects and disorders while the G band is assigned to the second ordering scattering of E_{2g} phonon of the $C\text{ sp}^2$ domains [39]. After thermal treatment at $500\text{ }^\circ\text{C}$ under flowing nitrogen, there is a slight increase in the integrated intensity ratios of the D and G bands (I_D/I_G) from 0.807 for GO to 0.816 for rGO, demonstrating that a more disordered carbon structure was produced after the thermal reduction of GO [40]. After decorating with CoO nanoparticles, the ratios of I_D/I_G became even larger, reaching 0.914 for rGO@Co-4, which can be ascribed to the induced stress and increased structural defects/disorders from the formed CoO nanoparticles on the rGO nanosheets and the thermal annealing process [39].

The chemical structures of GO, rGO and rGO@Co composites were characterized by FT-IR spectroscopy (Fig. 4). The characteristic absorption peaks at around 1049 , 1394 , and 1627 cm^{-1} correspond to the stretching vibration of C–O, O–H deformation, and the adsorbed H_2O

deformation respectively, while the weak absorption peaks near 1725 cm^{-1} is attributed to $\text{C}=\text{O}$ in carboxylic acid moieties [35]. Nevertheless, in the FT-IR spectrum of rGO@Co obtained by the thermal reduction method, few characteristic absorption peaks of C–O, C=O and O–H groups were observed, indicating that GO was thermally reduced to rGO. Furthermore, it is worth noting that after reduction of GO, two characteristic absorption peaks appeared at 2856 and 2921 cm^{-1} , which were attributed to the $-\text{CH}_2$ and $-\text{CH}_3$ groups on the graphene defects or edges, respectively [41]. Moreover, there are no obvious changes in the characteristic absorption peaks of the rGO@Co carriers with the increase of CoO content [42].

3.2. Morphology and structure of rGO@Co/PEG composite PCMs

Fig. 5 shows the SEM morphology of rGO/PEG and rGO@Co-6/PEG. It can be seen that the morphology of rGO/PEG is similar to that of rGO, indicating that the core material PEG was successfully adsorbed into the network structure of rGO nanosheets. However, the smooth structure of PEG was also observed on the sample surface and no CoO nanoparticles were observed on the surface in rGO@Co-6/PEG, indicating that PEG completely covered the support material.

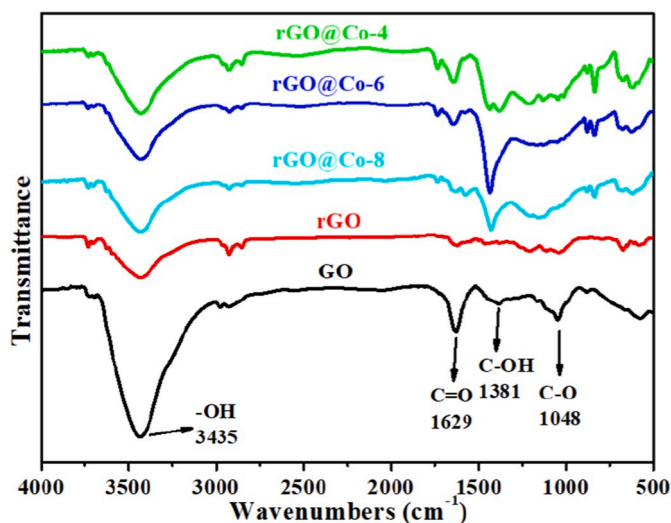


Fig. 4. FT-IR spectrum of the GO, rGO and rGO@Co composites.

In order to prove that the rGO@Co carriers successfully adsorbed PEG2000, the XRD patterns of the composite PCMs were characterized. Compared with the XRD patterns of rGO@Co-6, the typical diffraction peaks of PEG crystals were observed in rGO@Co-6/PEG at 19° and 23° (Fig. 6a), demonstrating that the crystal structure of PEG is well kept in the preparing process of rGO@Co/PEG composite. However, the diffraction peaks of CoO nanoparticles almost disappeared and the relative intensity of the broad diffraction peak of rGO at 15°–30° decreased after enwrapped by PEG, which can be ascribed to the high

content of PEG and the low amount of CoO in the rGO@Co-6/PEG composite. Similar results were also observed in the XRD patterns of other composite PCMs (Fig. S2). In the FT-IR analysis as shown in Fig. 6b and Fig. S3, all the characteristic peaks of PEG2000 can be found in the rGO@Co-6/PEG composite PCM, which proves that there is only simple physical action between PEG2000 and rGO@Co support without producing new structures, and the PEG2000 has been successfully loaded into the rGO@Co-6 support. The leakage test results of PEG2000, rGO/PEG, and rGO@Co/PEG composites are shown in the digital photos of Fig. 7. For pure PEG2000, when the system temperatures (e.g., 60 and 85 °C) were higher than its melting point (55.8 °C), PEG has been partially or completely melted into the liquid state through the solid state. For rGO/PEG and rGO@Co/PEG composites with around 80 wt% PEG loading, when the system temperatures (e.g., 60 and 85 °C) were higher than the melting point of PEG, no liquid PEG was observed on the surface of the composite PCMs, indicating that the prepared rGO/PEG and rGO@Co/PEG composites all have the good form-stability.

Thermal stability is another important indicator of composite PCMs. The TGA curves of rGO and rGO@Co supports are shown in Fig. S4. The weight loss from room temperature to around 140 °C for rGO@Co composite can be attributed to the removal of adsorbed H₂O/gases by CoO nanoparticles and rGO nanosheets. The supports have a high thermal stability up to 600 °C and have great potentials in improving the stability of composite PCMs. In this work, the supports were vacuumed at 100 °C for 24 h to remove the possible gases/H₂O before impregnating the PEG2000. After mixing with melted PEG2000, the composite PCMs were also dried at 80 °C for 24 h in vacuum to impregnate the PEG into rGO and rGO@Co supports. As shown in Fig. 8, PEG2000 starts weight loss at 350 °C and almost finishes at around 420 °C, which can be attributed to the breakage of PEG molecular chain. Nevertheless, the presence of trace residual mass after pyrolysis can be attributed to trace

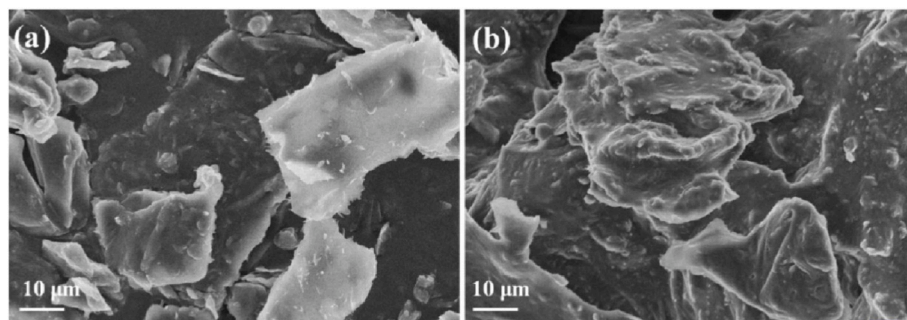


Fig. 5. SEM curves of rGO/PEG (a) and rGO@Co-6/PEG (b).

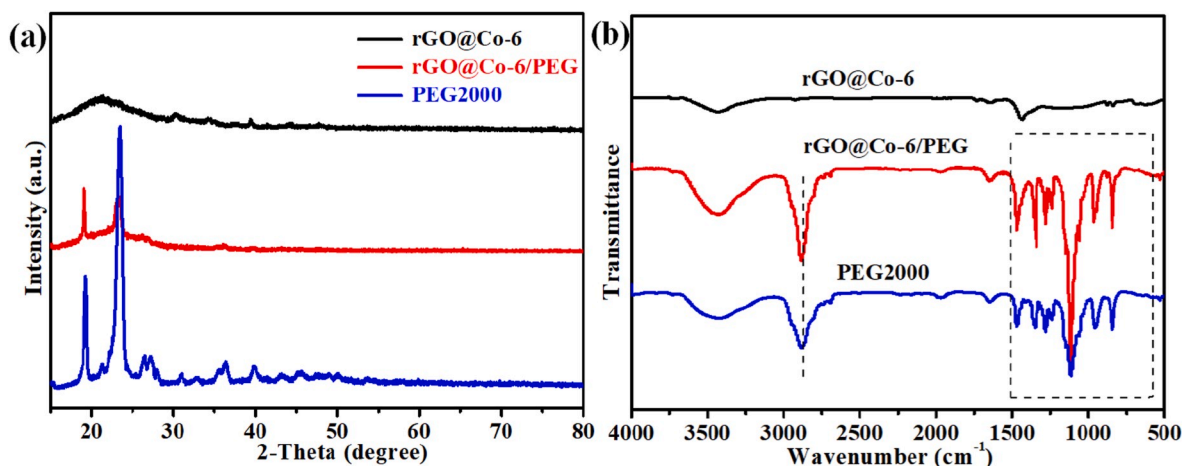


Fig. 6. XRD patterns (a) and FT-IR spectra (b) of PEG2000, rGO@Co-6 and rGO@Co-6/PEG.

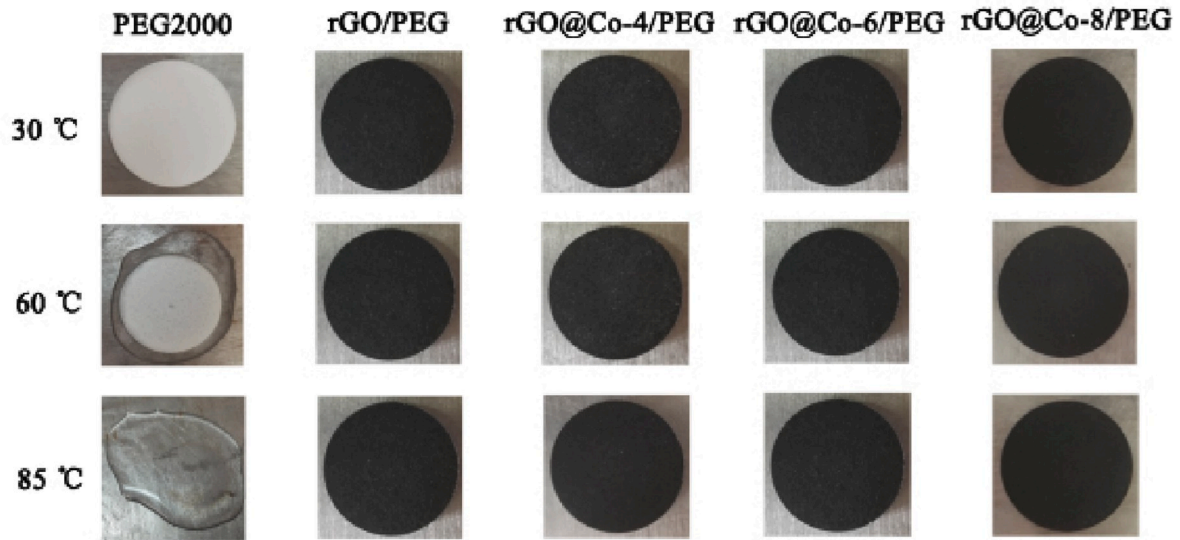


Fig. 7. Digital photos of PEG 2000, rGO/PEG, and rGO@Co/PEG composites at different temperature.

impurity or some residuals which need to be decomposed at higher temperature. As for composite PCMs, no obvious weight loss was observed in the temperature range of 40–300 °C, indicating that the composite PCMs can maintain excellent thermal stability at temperature lower than 300 °C. With the increasing temperature, the weight loss processes were carried out by only one step, similar to that of pure PEG2000, indicating the weight loss mainly coming from the decomposition of PEG molecular chain. The PEG contents in pure PEG and composite PCMs were determined based on their weight loss at 500 °C in Fig. 8. As summarized in Table 1, the actual weight losses for composite PCMs are slightly different with the theoretical loading of PEG (80%), which may be due to the uneven distribution of PEG or the increased decomposition temperature for those PEG confined within the rGO@Co.

3.3. Thermal performance of rGO@Co/PEG composite

Phase change behaviors of PEG2000, rGO/PEG and rGO@Co/PEG composites were measured by DSC, as shown in Fig. 9. The corresponding phase change enthalpies (H_M and H_S), melting temperature (T_M) and solidifying temperature (T_S) are summarized in Table 1. It is observed that the enthalpies of the prepared composite PCMs are significantly reduced and the melting temperatures are slightly

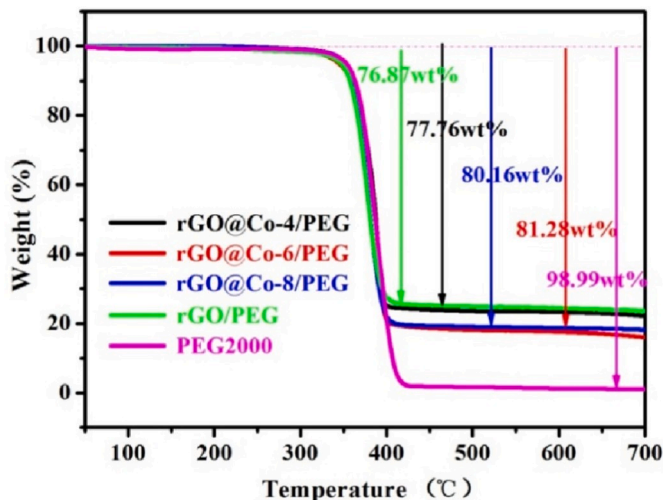


Fig. 8. TG curves of PEG2000, rGO/PEG and rGO@Co/PEG.

Table 1

DSC data of PEG2000, rGO/PEG, rGO@Co/PEG composite PCMs.

Samples	PEG contents	Melting process		Solidifying process	
		T_M (°C)	H_M (J/g)	T_S (°C)	H_S (J/g)
PEG2000	98.99%	55.81	164.93	17.62	149.38
rGO/PEG	76.87%	58.28	121.34	19.78	115.24
rGO@Co-8/PEG	80.16%	61.50	131.94	19.89	122.63
rGO@Co-6/PEG	81.28%	56.58	130.29	20.41	123.56
rGO@Co-4/PEG	77.76%	55.82	128.65	22.72	116.64

increased compared to pure PEG. The phenomenon can be attributed to the more serious obstacle effect of the microporous structure of the carriers on the thermal diffusion motion of the PEG molecular segments. In addition, compared with pure PEG, the slightly higher solidification temperature of composite PCMs can be attributed to the provided nucleation sites from porous supports for PEG crystallization during the solidification process, which accelerates the PCM crystallization. Furthermore, as shown in Fig. S6, the degree of supercooling of pure PEG and complexes increases first and then decreases with increasing CoO content, which can be attributed to the following reasons: when there is no CoO or low CoO content in the composite (e.g., rGO/PEG or rGO@Co-8/PEG), the rGO or rGO@Co network will hinder the thermal diffusion movement of the PEG molecular segment and limit the growth of PEG crystal grains, so the degree of supercooling will increase. When the CoO content increased (e.g., rGO@Co-6/PEG or rGO@Co-4/PEG), the CoO nanoparticles on the surface of rGO can produce a heterogeneous nucleation effect and increase the number of nucleation points in the polymer matrix, which is conducive to reducing the degree of supercooling. In order to reveal in detail the phase transition confined behavior of PCM molecules in the carriers, the corresponding crystallinity (F_c) was calculated by Eq. (1):

$$F_c = \frac{\Delta H_{PCM}}{\Delta H_{pure}\beta} \times 100\% \quad (1)$$

where ΔH_{pure} and ΔH_{PCM} are the latent heat of pure PCM and composite PCMs respectively, and β represents the mass fraction of PCM molecules in composite PCMs. The F_c can effectively reflect the interaction between PCM and supports. As summarized in Table 1, the melting enthalpy (H_M) of rGO/PEG, rGO@Co-8/PEG, rGO@Co-6/PEG, and rGO@Co-4/PEG composites reached 121.34, 131.94, 130.29, and 128.65 J/g respectively, and their F_c were 92%, 99.8%, 98.7%, and

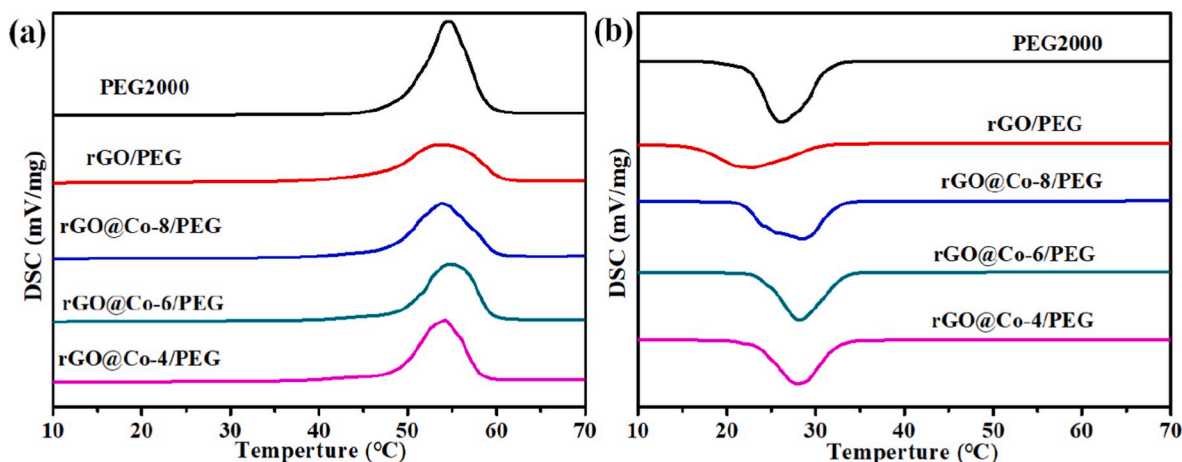


Fig. 9. DSC curves of PEG2000, rGO/PEG and rGO@Co/PEG during heating (a) and cooling (b).

97.5%. Based on the above analysis results, it can be concluded that rGO@Co/PEG composite PCMs with high thermal energy storage capacities and efficiencies have promising potentials in practical applications. As characterized by DSC (Fig. 10a), XRD (Fig. 10b), SEM (Fig. S5) and FT-IR (Fig. S7), the composite PCMs have good cycle stability for at least 50 cycles.

Thermal conductivity is another important parameter to be considered in solar-to-thermal conversion and thermal energy storage applications of composite PCMs. As shown in Fig. 11, the low thermal conductivity (0.2072 W/mK) of pure PEG2000 in the solid state indicates its low solar-to-thermal conversion and thermal energy storage/retrieval efficiency. Accordingly, after using rGO as PEG support, the thermal conductivity for PCM-0 was measured to 0.3145 W/mK due to the relatively higher thermal conductivity of rGO itself [43]. With the increased CoO amount in the rGO@Co support, the thermal conductivities increased to 0.3751 W/mK for PCM-8 and 0.4521 W/mK for PCM-6. Furthermore, the thermal conductivity of PCM-4 is increased to 0.5231 W/mK, which has 152.46% and 66.33% improvement compared with those of pure PEG2000 and PCM-0 respectively maybe owing to the high thermal conductivity of CoO nanoparticles [44]. The thermal conductivity improvement also demonstrates that the CoO nanoparticles embedded in/on the surface of rGO nanosheets provide multiple thermal conductive pathways as well as accelerate the thermal transport between rGO and PEG [45].

The solar-to-thermal conversion performance of as-prepared composite PCMs were investigated and the measurement schematic diagram is shown in Fig. 12a. The samples were placed under simulated sunlight

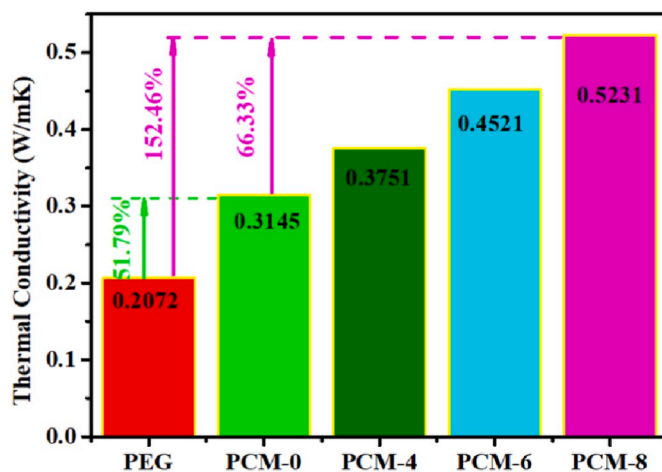


Fig. 11. Thermal conductivities of PEG2000, rGO/PEG, rGO@Co/PEG composites.

using a 300 W xenon lamp and the temperature change was monitored by the infrared temperature probe and recorded by the data acquisition system with Temp Monitor S2 software. The temperature of the samples increased after solar radiation, as shown in Fig. 12b. When the temperature exceeded 45 °C, rGO/PEG, rGO@Co/PEG composites warmed up faster than pure PEG2000. When the temperature rose to the melting

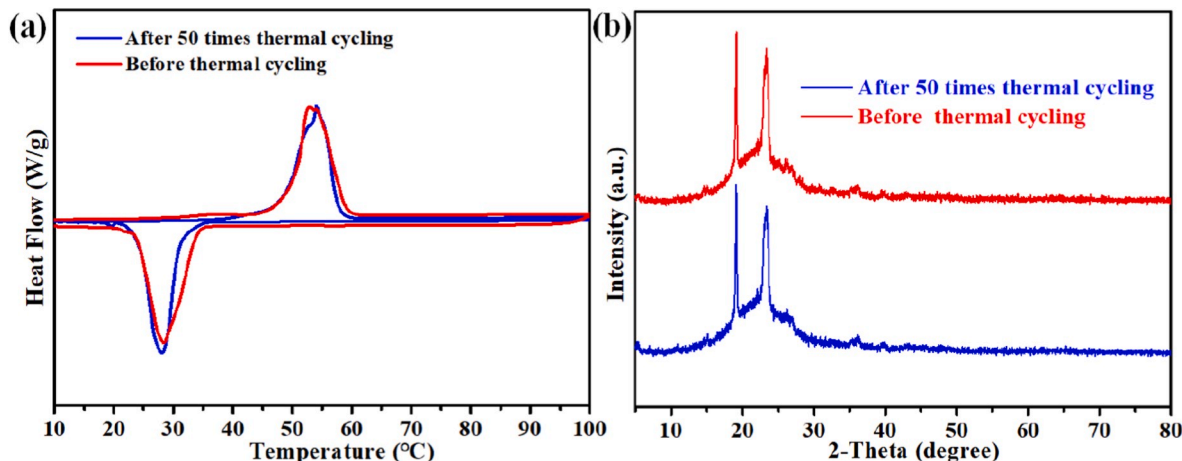


Fig. 10. DSC curves (a) and XRD patterns (a) of rGO@Co-6/PEG composite PCMs before and after 50 thermal cycling.

point of PEG2000, the curve for PEG was relatively gentle while the curves for rGO@Co/PEG composites showed an obvious inflection point, indicating that organic molecules changed from solid phase to liquid phase. Continuing to radiate under the simulated sunlight, the curve for rGO@Co-6/PEG composite reached a maximum temperature at about 75.8 °C within 450 s while a temperature platform appeared in the curve for pure PEG, indicating the balance between solar energy absorption and heat dissipation to the surrounding environment. By comparing the highest temperatures reached between pure PEG and the composite within 450 s, it is concluded that the order of solar-to-thermal conversion efficiency is $\text{rGO@Co-6/PEG} > \text{rGO@Co-4/PEG} > \text{rGO@Co-8/PEG} > \text{rGO/PEG} > \text{pure PEG}$, and the CoO nanoparticles in rGO@Co supports can improve the solar-thermal conversion ability of rGO. When the solar radiation is eliminated, the temperatures of the composite PCMs drop rapidly, especially for rGO@Co/PEG composites, which means that the complex released a large amount of heat during solidification process. As we all know, there are three main ways of thermal transfer: thermal conduction, thermal convection and thermal radiation [46]. During the solar-to-thermal conversion process, the thermal transfer from the sun to the surface of the composite PCMs is thermal radiation, and the thermal transfer within the solid materials is thermal conduction. Therefore, thermal conductivity will be the main factor determining the solar-to-thermal conversion efficiency of solid composites. The above analysis shows that the solar-to-thermal conversion efficiencies of the rGO/PEG and rGO@Co/PEG composites are much higher than that of pure PEG, which can be due to the significantly superior thermal conductivity and light absorption ability of rGO and CoO than pure PEG. In addition, the rGO@Co/PEG composites with CoO nanoparticles exhibit higher light absorption ability than rGO/PEG and increases with the CoO amount [47], as determined by the UV-vis spectrum of rGO/PEG and rGO@Co/PEG composites (Fig. S8). Nevertheless, the solar-to-thermal conversion capacity of rGO@Co-6/PEG is slightly higher than that of rGO@Co-4/PEG, which may be due to the increased interfacial thermal resistance and hindered phonon transmission resulted from the uneven size distribution and aggregation of CoO nanoparticles in rGO@Co-4 since the filling amount, shape and size of thermally conductive fillers have an impact on the solar-thermal energy conversion and thermal energy transfer [48].

Combined with the characterization results and thermal properties of rGO@Co and rGO@Co/PEG PCMs, a possible mechanism for the enhanced solar-to-thermal conversion and storage has been proposed, as shown in Scheme 2. The functional area of the solar-to-thermal conversion in PCMs is mainly divided into three parts: light harvesting part, solar-to-thermal conversion part and thermal energy storage part. In the rGO@Co/PEG composite PCMs under the irradiation of simulated sunlight, rGO nanosheets and CoO nanoparticles with good light absorption

ability absorb the solar energy to become solar excited states (Step A) [47]. During the solar-to-thermal conversion part, the solar energy in the solar excited rGO nanosheets and CoO nanoparticles was converted into thermal energy, which can be transferred by rGO and CoO nanoparticles to their contacted PEG chains with higher enthalpy capacity (Step B). Moreover, as demonstrated by the SEM and TEM results, CoO nanoparticles are well embedded into the rGO nanosheets while part of the CoO nanoparticles are also exposed to the atmosphere. Therefore, during the thermal energy storage part, the highly thermally conductive CoO nanoparticles can act as the thermal energy transfer intermediates or bridges between encapsulated PEG and rGO nanosheets, which would reduce the interface thermal resistance between the carriers and PEG and improve the thermal storage efficiency of the composites. When there is no continuing solar radiation, the rGO@Co/PEG composites with high enthalpy energy can release the stored thermal energy for special applications (Step C).

4. Conclusions

The rGO@Co composites with well-distributed CoO nanoparticles and large porosity were synthesized via a facile thermal annealing method and applied as PEG support for solar-to-thermal energy conversion and storage. Our results show that the rGO@Co/PEG composite PCMs can effectively store the thermal energy coming from solar energy absorption and conversion. The high thermal conductivity and high light adsorption ability of CoO nanoparticles and rGO nanosheets and their intimate contact improved the thermal conductivity with 51.79%–152.46% improvement of pure PEG and solar-to-thermal conversion efficiency of composite PCMs. In addition, the obtained rGO@Co/PEG composite PCMs have high energy storage capacities (131.94 J/g) and thermal energy storage/release rates, and outstanding form stability for at least 50 cycles, which could offer promising applications for solar energy harvesting, conversion, and storage as thermal energy.

Declaration of competing interest

1We confirm that there are no interests to declare associated with this publication.

2We confirm that all of the sources of funding for the work described in this publication are acknowledged.

CRediT authorship contribution statement

Yang Li: Validation, Investigation, Formal analysis, Data curation, Writing - original draft. **Yaqiong Li:** Methodology, Validation, Investigation, Formal analysis, Writing - original draft. **Xiubing Huang:**

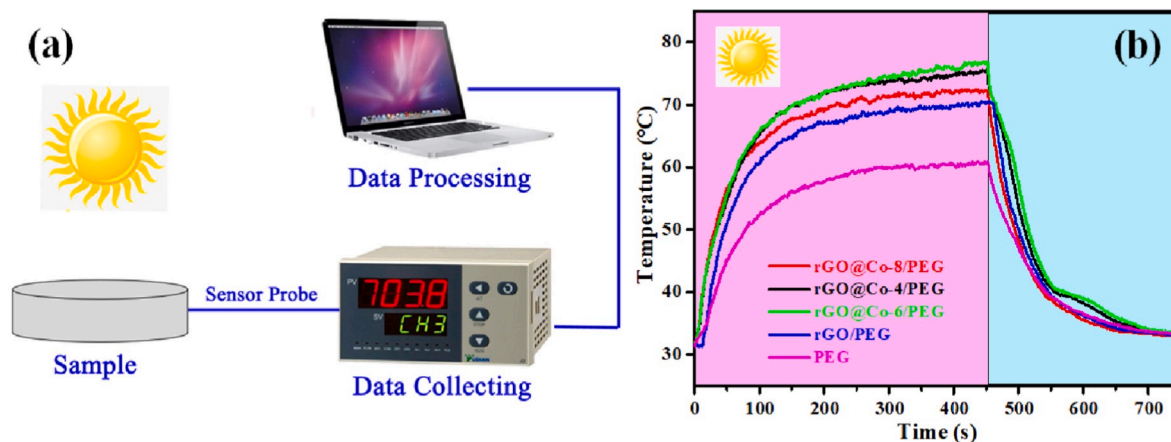
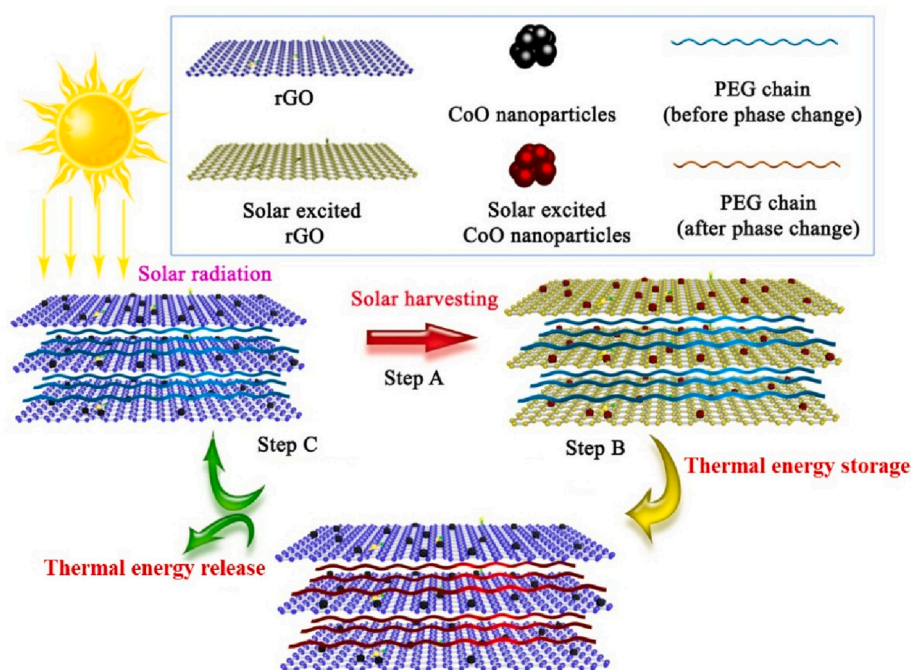


Fig. 12. The solar-to-thermal energy conversion and storage results: (a) the schematic illustration of solar-to-thermal conversion test and (b) the solar-to-thermal conversion curves.



Scheme 2. Solar-to-thermal energy conversion and storage mechanism in rGO/Co/PEG composites.

Conceptualization, Supervision, Writing - review & editing, Funding acquisition. **Haiyan Zheng:** Resources, Data curation. **Guilong Lu:** Resources, Data curation. **Zuoshuai Xi:** Resources, Data curation. **Ge Wang:** Supervision, Project administration, Writing - review & editing, Funding acquisition.

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Appendix A. Supplementary data

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